12.434

EE25BTECH11004 - Aditya Appana

October 12, 2025

Question

The two sides of a parallelogram are given by the vectors $\mathbf{A} = 2\hat{\imath} - 3\hat{\jmath}$ and $\mathbf{B} = 3\hat{\imath} + 2\hat{\jmath}$. The area of the parallelogram is:

Solution

Let:

$$\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix} \tag{1}$$

$$\mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \tag{2}$$

The area of the parallelogram can be expressed as:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} |\mathbf{A_{23} B_{23}}| \\ |\mathbf{A_{31} B_{31}}| \\ |\mathbf{A_{12} B_{12}}| \end{pmatrix} \tag{3}$$

Substituting values:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 13 \end{pmatrix} \tag{4}$$

Therefore, the area of the parallelogram is:

$$||\mathbf{a} \times \mathbf{b}|| = 13 \tag{5}$$

It can be seen that $\mathbf{a}^T\mathbf{b} = 0$ and $||\mathbf{a}|| = ||\mathbf{b}||$, therefore they form the sides of a square.
Therefore the area can also be calculated as:

$$||\mathbf{a}||^2 = \sqrt{13}^2 = 13 \tag{6}$$

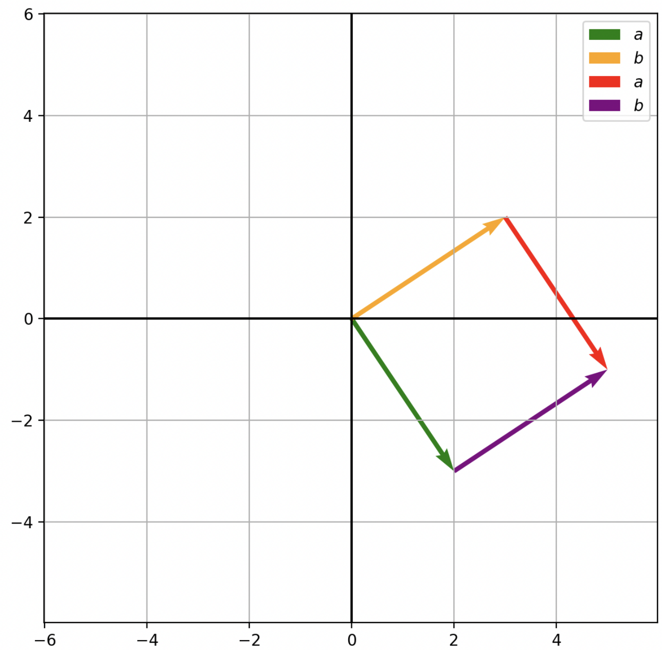

Figure 1: Plot